Exercise 09a

Let γ be the opening operator and φ the closing operator
Define:

$$F(I) = \varphi(\gamma(I)).$$

We prove idempotence by showing:

$$F(F(I)) \leq F(I) \quad \textit{and} \quad F(F(I)) \geq F(I).$$

We know that

$$F(F(I)) = \varphi\big(\gamma(\varphi(\gamma(I)))\big).$$

1. Prove $F(F(I)) \geq F(I)$

Since closing is extensive:

$$\gamma(I) \leq \varphi(\gamma(I)).$$

Because opening is increasing:

$$\gamma(\gamma(I)) \leq \gamma(\varphi(\gamma(I))).$$

Using idempotence of opening:

$$\gamma(I) \leq \gamma(\varphi(\gamma(I))).$$

Now apply φ (increasing):

$$\varphi(\gamma(I)) \leq \varphi(\gamma(\varphi(\gamma(I)))).$$

Therefore,

$$F(I) \leq F(F(I)).$$

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2. Prove $F(F(I)) \leq F(I)$

Since opening is anti-extensive:

$$\gamma(\varphi(\gamma(I))) \leq \varphi(\gamma(I)).$$

Applying φ (increasing):

$$\varphi(\gamma(\varphi(\gamma(I)))) \leq \varphi(\varphi(\gamma(I))).$$

Using idempotence of closing, we replace this equal on the equation, cause we know that applying closing two times, it is the same that applying once:

$$\varphi(\varphi(\gamma(I))) = \varphi(\gamma(I)).$$

And replacing our F function

$$F(F(I)) \leq F(I).$$

Since

$$F(F(I)) \leq F(I) \quad and \quad F(F(I)) \geq F(I),$$

we conclude

$$F(F(I)) = F(I).$$

Therefore, the closing–opening alternated filter is idempotent.